

- RAG Learning

## 1.4 Powers, Roots, Surds & Standard Form

AQA GCSE Mathematics 8300 — Number — spec refs N6–N9 — Papers 1 (non-calculator), 2 and 3 (calculator)

### 1.4.1 Powers and Roots

A **power** (or index) tells you how many times a number is multiplied by itself:  $a^n = a \times a \times \dots \times a$  ( $n$  times). In  $5^3$ , 5 is the **base** and 3 is the **index**.

A **root** is the inverse of a power:  $\sqrt{49} = 7$  because  $7^2 = 49$ ;  $\sqrt[3]{64} = 4$  because  $4^3 = 64$ ;  $\sqrt[4]{81} = 3$  because  $3^4 = 81$ .

Every positive number has two square roots, one positive and one negative ( $\pm 7$  for 49), but the  $\sqrt{\phantom{x}}$  symbol means the positive one. Cube roots of negative numbers are negative:  $\sqrt[3]{-8} = -2$ .

### Numbers You Should Know

| Type              | Values                                                        |
|-------------------|---------------------------------------------------------------|
| Squares to $15^2$ | 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225 |
| Cubes             | 1, 8, 27, 64, 125 (and $10^3 = 1000$ )                        |
| Powers of 2       | 2, 4, 8, 16, 32, 64, 128, 256                                 |
| Powers of 3       | 3, 9, 27, 81, 243                                             |
| Powers of 4       | 4, 16, 64, 256                                                |
| Powers of 5       | 5, 25, 125, 625                                               |

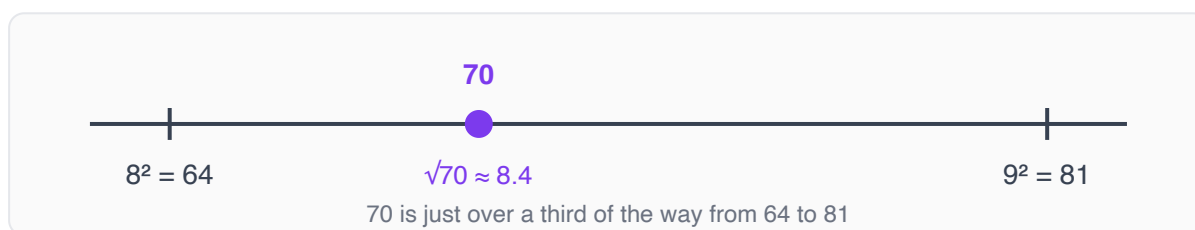
Powers of 10

 $10^3 = 1000$  (one thousand),  $10^6 = 1\,000\,000$  (one million)**Worked Example:** Work out  $2^3 \times 3^2 - \sqrt{144}$ .

$$8 \times 9 - 12 = 72 - 12 = 60$$

## Estimating Powers and Roots Higher

To estimate a root without a calculator, find the square (or cube) numbers either side, then refine.

**Worked Example:** Estimate  $\sqrt{70}$  to 1 decimal place.

$$8^2 = 64 \text{ and } 9^2 = 81, \text{ so } 8 < \sqrt{70} < 9.$$

$$8.3^2 = 68.89 \text{ and } 8.4^2 = 70.56. 70 \text{ is closer to } 70.56, \text{ so } \sqrt{70} \approx 8.4$$

**Worked Example:** Estimate  $2.9^4$ .

$$2.9 \approx 3, \text{ so } 2.9^4 \approx 3^4 = 81 \text{ (a slight overestimate, as } 2.9 < 3).$$

## 1.4.2 Laws of Indices

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$a^0 = 1 \quad a^{-n} = \frac{1}{a^n} \quad a^1 = a$$

The laws only work when the **bases are the same**.

**Worked Example:** Simplify  $4^7 \times 4^{-2} \div 4^3$ , giving your answer as a power of 4.

$$4^{7+(-2)-3} = 4^2 \text{ (which equals 16)}$$

**Worked Example:** Simplify  $(2x^3)^4$ .

Raise every part of the bracket to the power 4:  $2^4 \times x^{12} = 16x^{12}$

## Zero and Negative Indices

Why is  $a^0 = 1$ ? By the division law,  $5^3 \div 5^3 = 5^0$ ; but any non-zero number divided by itself is 1, so  $5^0 = 1$ .

Similarly  $5^2 \div 5^4 = 5^{-2}$ , and  $\frac{25}{625} = \frac{1}{25}$ , so  $5^{-2} = \frac{1}{5^2}$ . A negative index means **reciprocal**.

**Worked Example:** Work out (a)  $2^{-3}$  (b)  $\left(\frac{3}{4}\right)^{-2}$

$$(a) \frac{1}{2^3} = \frac{1}{8} \quad (b) \text{ flip the fraction: } \left(\frac{4}{3}\right)^2 = \frac{16}{9}$$

**Worked Example:** Find  $n$  when  $3^n = 81 \times 3^2$ .

$$81 = 3^4, \text{ so } 3^n = 3^4 \times 3^2 = 3^6, \text{ giving } n = 6$$

**Exam Tip:** A negative index does **not** make the answer negative:  $2^{-3} = \frac{1}{8}$ , not  $-8$ . And  $(-2)^2 = 4$  but  $-2^2 = -4$  — brackets matter.

## Fractional Indices

Higher

$$a^{\frac{1}{n}} = \sqrt[n]{a} \qquad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

The **denominator** is the root, the **numerator** is the power, and a **minus sign** means reciprocal.

**Worked Example:** Evaluate  $8^{\frac{2}{3}}$ .

Root first:  $\sqrt[3]{8} = 2$ . Then power:  $2^2 = 4$

**Worked Example:** Evaluate  $16^{-\frac{1}{2}}$ .

Reciprocal:  $\frac{1}{16^{\frac{1}{2}}} = \frac{1}{\sqrt{16}} = \frac{1}{4}$

**Worked Example:** Evaluate  $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$ .

Reciprocal:  $\left(\frac{8}{27}\right)^{\frac{2}{3}}$ . Cube root:  $\frac{2}{3}$ . Square:  $\frac{4}{9}$

**Worked Example:** Solve  $9^x = \frac{1}{27}$ .

Write both sides as powers of 3:  $3^{2x} = 3^{-3}$ , so  $2x = -3$  and  $x = -\frac{3}{2}$

**Exam Tip:** With a fractional index, take the root *first* — the numbers stay small.  $64^{\frac{2}{3}} = 4^2 = 16$  is much easier than  $\sqrt[3]{4096}$ .

### 1.4.3 Calculating Exactly with Multiples of $\pi$

"Give your answer in terms of  $\pi$ " means leave  $\pi$  in the answer as a symbol. This gives an **exact** answer instead of a rounded decimal.

**Worked Example:** A circle has diameter 10 cm. Find its exact area and circumference.

$$r = 5: \text{Area} = \pi \times 5^2 = \mathbf{25\pi \text{ cm}^2}; \text{Circumference} = \pi \times 10 = \mathbf{10\pi \text{ cm}}$$

**Worked Example:** A circle of diameter 10 cm is cut from a square of side 10 cm. Find the exact area left.

$100 - 25\pi \text{ cm}^2$  — this cannot be simplified further, because 100 and  $25\pi$  are not like terms.

**Exam Tip:** Treat  $\pi$  like a letter in algebra:  $3\pi + 5\pi = 8\pi$ ,  $2 \times 6\pi = 12\pi$ , but  $18 + 3\pi$  stays as it is.

## 1.4.4 Surds Higher

A **surd** is a root that cannot be written exactly as a fraction or terminating decimal, e.g.  $\sqrt{2}$ ,  $\sqrt{5}$ ,  $\sqrt{12}$ . Leaving answers in surd form keeps them **exact**. ( $\sqrt{9} = 3$  is not a surd.)

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad \sqrt{a} \times \sqrt{a} = a$$

But  $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ .

### Simplifying Surds

Find the **largest square factor** of the number under the root.

**Worked Example:** Simplify  $\sqrt{12}$  and  $\sqrt{72}$ .

$$\sqrt{12} = \sqrt{4 \times 3} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$$

$$\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$$

**Worked Example:** Simplify  $\sqrt{75} - \sqrt{27} + \sqrt{12}$ .

$$5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3} = 4\sqrt{3} \quad (\text{collect like surds just like like terms})$$

## Expanding Brackets with Surds

**Worked Example:** Expand and simplify  $(2 + \sqrt{3})^2$ .

$$(2 + \sqrt{3})(2 + \sqrt{3}) = 4 + 2\sqrt{3} + 2\sqrt{3} + 3 = 7 + 4\sqrt{3}$$

**Worked Example:** Expand and simplify  $(3 + \sqrt{2})(3 - \sqrt{2})$ .

$$9 - 3\sqrt{2} + 3\sqrt{2} - 2 = 7 \quad (\text{difference of two squares: } a^2 - b)$$

## Rationalising the Denominator

To **rationalise** means to remove the surd from the denominator of a fraction.

$$\frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{b}$$

$$\frac{a}{c + \sqrt{b}} : \text{multiply top and bottom by } c - \sqrt{b}, \text{ so the denominator becomes } c^2 - b$$

**Worked Example:** Rationalise  $\frac{6}{\sqrt{3}}$ .

$$\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$$

**Worked Example:** Rationalise  $\frac{7}{3 + \sqrt{2}}$ .

$$\frac{7(3 - \sqrt{2})}{(3 + \sqrt{2})(3 - \sqrt{2})} = \frac{21 - 7\sqrt{2}}{9 - 2} = \frac{21 - 7\sqrt{2}}{7} = 3 - \sqrt{2}$$

**Exam Tip:** "Give your answer in the form  $a\sqrt{b}$ " means  $b$  must be as small as possible:  
 $\sqrt{72} = 6\sqrt{2}$ , not  $2\sqrt{18}$  or  $3\sqrt{8}$ .

## 1.4.5 Standard Form

A number is in **standard form** when it is written as

$$A \times 10^n \quad \text{where } 1 \leq A < 10 \text{ and } n \text{ is an integer}$$

Large numbers have a positive power; numbers between 0 and 1 have a negative power.

Large number

45 000

point moves 4 places

$$4.5 \times 10^4$$

Small number

0.00062

point moves 4 places

$$6.2 \times 10^{-4}$$

| Ordinary number | Standard form     |
|-----------------|-------------------|
| 320 000         | $3.2 \times 10^5$ |

|            |                       |
|------------|-----------------------|
| 56 000 000 | $5.6 \times 10^7$     |
| 0.0007     | $7 \times 10^{-4}$    |
| 0.0000362  | $3.62 \times 10^{-5}$ |

## Multiplying and Dividing

Deal with the number parts and the powers of 10 separately, then adjust so that  $1 \leq A < 10$ .

**Worked Example:** Work out  $(6 \times 10^4) \times (5 \times 10^3)$ .

$6 \times 5 = 30$  and  $10^4 \times 10^3 = 10^7$ , giving  $30 \times 10^7$ .

30 is not between 1 and 10:  $30 \times 10^7 = 3 \times 10^1 \times 10^7 = \mathbf{3 \times 10^8}$

**Worked Example:** Work out  $(8.4 \times 10^6) \div (2.1 \times 10^{-2})$ .

$8.4 \div 2.1 = 4$  and  $10^6 \div 10^{-2} = 10^{6-(-2)} = 10^8$ , so the answer is  $\mathbf{4 \times 10^8}$

## Adding and Subtracting

The powers must match, so convert to ordinary numbers (or to the same power of 10) first.

**Worked Example:** Work out  $(4.2 \times 10^5) - (3.5 \times 10^4)$ .

$420\,000 - 35\,000 = 385\,000 = \mathbf{3.85 \times 10^5}$

## Calculator Displays

Calculators may show standard form as 4.7E-3 or  $4.7 \times 10^{-3}$ . Both mean  $4.7 \times 10^{-3} = 0.0047$ .

Always rewrite these properly; "4.7E-3" is not accepted as an answer.

**Worked Example:** Light travels at  $3 \times 10^5$  km/s. The Sun is  $1.5 \times 10^8$  km from the Earth. How long does sunlight take to reach the Earth?

$$\text{time} = \frac{1.5 \times 10^8}{3 \times 10^5} = 0.5 \times 10^3 = 500 \text{ seconds} = \mathbf{8 \text{ minutes } 20 \text{ seconds}}$$

**Exam Tip:** After multiplying, check that  $A$  is between 1 and 10. Answers such as  $30 \times 10^7$  or  $0.5 \times 10^3$  are correct in value but are **not** in standard form and will lose the final mark.

## Summary

| Topic                        | Key fact                                                                                                  |
|------------------------------|-----------------------------------------------------------------------------------------------------------|
| Index laws                   | Multiply $\rightarrow$ add powers; divide $\rightarrow$ subtract; power of a power $\rightarrow$ multiply |
| Zero / negative index        | $a^0 = 1$ , $a^{-n} = 1/a^n$                                                                              |
| Fractional index<br>(Higher) | $a^{m/n} = (\sqrt[n]{a})^m$                                                                               |
| Exact answers                | Leave $\pi$ and surds in the answer                                                                       |
| Surds (Higher)               | Simplify using square factors; rationalise by multiplying by $\sqrt{b}$ or the conjugate                  |
| Standard form                | $A \times 10^n$ , $1 \leq A < 10$ , $n$ an integer                                                        |